

# Cable Coil and Stow SOP (1x Episode: 4 Cables)

One episode coils, straps, and stows **four cables** using the two-arm tabletop robot. The table begins with four loose cables on the left, a clear shared work area in the center, and a strap basket plus an empty four-compartment organizer on the right. Cables are worked one at a time, and the organizer fills from left to right, so the episode ends with one finished bundle in each of the four compartments and the left input zone empty.

For each cable the workflow is: straighten the selected cable between the grippers, build a compact loop bundle in the air by alternating the grippers so that each one in turn holds the bundle while the other delivers the next loop, return the finished bundle to the left gripper, lay one hook-and-loop strap across the middle of the table, place the coiled cable on top of it, fasten the strap around the bundle middle, and place the secured bundle in the next empty organizer compartment. Only after a cable is stowed does the next cable leave the left input zone.

## Setup

Complete both checklists before recording starts.

### Cell configuration

- **Environment camera:** 900 mm.
- **cell\_type:** bimanual

### Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the full tabletop, all four loose cables on the left, the center work area, the strap basket, and the four-compartment organizer.
3. Both arms are at home with grippers open.
4. The table is clear except for the cables, strap basket, organizer, and robot hardware.

### Materials checklist

1. Stage four flexible cables loose and uncoiled in the left input zone. Their connectors are visible and the cables are not knotted together.
2. Place at least four reusable hook-and-loop straps in the shallow basket at the front-right of the table. Each strap begins open and separate from the cables.
3. Place the empty four-compartment organizer behind the strap basket in the back-right output zone. All four compartments are empty when recording starts.

4. Keep the shared center work area clear.
5. Before collection, confirm that every cable can form a compact loop bundle that fits freely in one organizer compartment and that one strap can overlap securely around its middle.

## Workspace layout

- **Left input zone:** loose cables that have not yet been completed
- **Center work area:** cable selection, straightening, airborne coiling, and fastening
- **Front-right supply zone:** basket of open hook-and-loop straps
- **Back-right output zone:** four-compartment organizer, filled left to right

## Arm assignments

- **Left gripper:** selects the cable, holds the starting section, holds the bundle for the first loop and every later odd-numbered loop, delivers the even-numbered loops, holds the finished bundle, and lowers it onto the strap for fastening.
- **Right gripper:** takes the far section to straighten the cable, delivers the first loop and every later odd-numbered loop, holds the bundle during the even-numbered loops, retrieves and lays out the strap, closes it, and carries the finished bundle to the organizer.

During Step 3 the two grippers swap the holding and delivering roles on every loop. Outside Step 3 the assignments are fixed: the right gripper straightens, stages and closes the strap, and stows; the left gripper holds and positions the bundle. The assignments are the same for all four cables.

## Vocabulary

- **Cable body:** the flexible insulated length between the connectors. Grasp the body, not a connector.
- **Starting end:** the cable end held first by the left gripper. The left gripper keeps the cable body just in from this end while the bundle is formed.
- **Free end:** the opposite end, reached last as the right gripper works along the cable.
- **Loop bundle:** a compact stack of same-direction loops gathered together in one gripper. The loops lie on top of one another; they do not alternate into a figure eight.
- **Holding gripper:** the gripper closed on the bundle for the current loop. It supports every loop gathered so far and receives the next delivered section.
- **Delivering gripper:** the other gripper for the current loop. It advances along the cable, grasps the next section, and brings it to the holding gripper.
- **Alternate the hands:** swap the holding and delivering roles after every loop, so the bundle passes from gripper to gripper as it grows. Alternating the *hands* does not alternate the *loop direction*: every loop is still formed in the same direction and stacked, never crossed into a figure eight.

- **Bundle middle:** the section pinched by the holding gripper where the stacked loops meet and where the strap wraps.
- **Even loops:** loops that are visually similar in size and form one compact bundle without a single loop projecting far beyond the others.
- **Cable turn:** one pass through Steps 1–5 for one cable. The episode contains four turns, and a turn is finished only when its bundle rests in the organizer.
- **Next empty compartment:** the first empty organizer compartment when counting from left to right. Turn 1 fills the leftmost compartment, turn 2 the second, turn 3 the third, and turn 4 the rightmost.
- **Secure strap:** one strap closed around the bundle middle with its hook-and-loop surfaces overlapped so the loops stay together when the bundle is carried, without crushing or sharply bending the cable.

## Steps

Run Steps 1–5 once per cable, four times in a row. A cable is selected, straightened, coiled, strapped, and stowed before the next cable is touched, so exactly one cable is out of the left input zone at any moment. After the fourth bundle is stowed, end the episode with Step 6.

### Step 1: Select and isolate one cable

**Goal:** one cable is separated from the loose group and controlled by both grippers; the remaining cables stay in the left input zone.

1. With the **left gripper**, grasp the selected cable body just in from one visible connector. This becomes the starting end.
2. Lift the starting end clear of the other cables without grasping or dragging the connector.
3. With the **right gripper**, grasp the cable body toward the free end when it becomes accessible.
4. Move the selected cable into the center work area while leaving the remaining cables on the left.
5. If another cable crosses over the selected cable, pause and use the right gripper to lift the crossing free before moving the selected cable farther. Do not pull the cables apart.

**Check:** only the selected cable extends into the center. If another cable moved with it, return that cable to the left input zone before continuing.

### Step 2: Straighten the selected cable

**Goal:** the selected cable extends between the grippers without a knot, closed loop, or cable crossing, and both connectors are visible.

1. Keep the **left gripper** on the cable body near the starting end.

2. With the **right gripper**, grasp the cable body farther toward the free end.
3. Move the grippers apart horizontally so the section between them becomes straight. Keep the connectors free rather than clamping them in either gripper.
4. Release the right gripper, move it farther toward the free end, regrasp the cable body, and pull the next section straight.
5. Repeat the release, advance, regrasp, and straighten motion until the free connector is reached and the full cable has been checked.
6. If a closed loop or crossing remains, open it with the right gripper and repeat the straightening pass. If the cable catches, release rather than pulling hard.

**Check:** both connectors are visible and the cable has no knot, closed loop, or section wrapped around another cable. If not, repeat Step 2 before coiling.

### **Step 3: Form a loop bundle in the air**

**Goal:** the full cable forms a compact stack of even, same-direction loops held above the table, built by alternating the holding and delivering grippers on every loop, and finished in the left gripper.

The grippers alternate. The left gripper holds the bundle for loop 1 and every later odd-numbered loop; the right gripper holds it for every even-numbered loop. The gripper that is not holding is the one that advances and delivers. The bundle stays in the air for the whole step, including through every hand-off.

#### **3.1 Set the starting end and the first loop**

1. With the **left gripper**, hold the cable body just in from the starting connector and raise it above the center work area. Let the connector hang freely. The left gripper is the holding gripper for the first loop.
2. With the **right gripper**, grasp the next section of cable far enough from the holding gripper to create one compartment-sized loop.
3. Bring that section through the air to the left gripper, forming a relaxed loop.
4. Slightly open the left gripper, place the delivered section against the section already held, and close the left gripper around both. Release the right gripper.

**Expected state:** one compartment-sized loop hangs from the left gripper above the center work area, both connectors are free, and the rest of the cable still runs down to the table.

#### **3.2 Hand the bundle over**

1. Bring the **delivering gripper** to the bundle middle beside the current grip and close it on the bundle.
2. Only then open the gripper that was holding and withdraw it. Grip before release, on every hand-off.

3. The two grippers have now swapped roles for the next loop. Keep the bundle above the center work area throughout the hand-off; do not let a loop slip free.

**Expected state:** every loop gathered so far is held at the bundle middle by the gripper that delivered the last loop, no loop has slipped free, and the bundle has not touched the tabletop.

### 3.3 Repeat the loop cycle to the free end

1. With the **delivering gripper**, move farther toward the free end, grasp the next equal-length section, and bring it to the holding gripper **in the same direction as every earlier loop**.
2. Slightly open the holding gripper, place the delivered section against the sections already held, and close the holding gripper around all of them. Release the delivering gripper.
3. Run 3.2 again after every loop, so the grippers swap on each one.
4. Repeat until the free connector reaches the bundle. Let the free connector rest beside the loops without pulling it tight.

**Expected state:** the whole cable is gathered into one stack of same-direction loops of similar size, held in the air, with the free connector resting against the loops.

### 3.4 Close the alternation in the left gripper

1. The last section must be delivered by the **right gripper**, so that the completed bundle is held by the **left gripper**.
2. If the free connector arrives while the right gripper is the holding gripper, close the alternation with one final 3.2 hand-off back to the left gripper before continuing.
3. Keep the completed bundle suspended in the left gripper until the right gripper has laid the strap across the middle of the table.

**Expected state:** the completed bundle hangs from the left gripper above the center work area, and the right gripper is empty and free to fetch a strap.

**Check:** the loops are stacked in one compact bundle, are visually similar in size, and fit within the outline of one organizer compartment. The grippers alternated on every loop, and the finished bundle is in the left gripper. If one loop is twisted, sharply bent, or much larger than the others, use the delivering gripper to remove and remake that loop before continuing.

## Step 4: Place the bundle on one strap and secure it

**Goal:** one open strap lies across the middle of the table, the coiled cable is placed on top of it, and the strap closes around the bundle middle.

### 4.1 Lay one strap across the middle of the table

1. Keep the **left gripper** closed on the bundle middle and hold the bundle above the center work area.
2. With the **right gripper**, take one open hook-and-loop strap from the front-right basket.

3. Bring the strap to the middle of the table and lay it open and flat across the center work area. Orient it so it will pass under the bundle middle, with both ends accessible for fastening.

**Expected state:** one open strap lies flat across the middle of the table, the remaining straps stay in the basket, and the bundle is still suspended in the left gripper above the center work area.

#### **4.2 Place the coiled cable on top of the strap**

1. With the **left gripper**, lower the coiled cable directly onto the open strap.
2. Center the bundle middle over the strap so the strap runs beneath all of the stacked loops.
3. Keep the bundle stable while leaving both strap ends exposed. Do not place the bundle on the bare tabletop and do not try to slide the strap underneath it afterward.

**Expected state:** the coiled cable rests on top of the strap in the middle of the table, the bundle middle is centered over the strap, and both strap ends are accessible.

#### **4.3 Close the strap around the bundle middle**

1. With the **right gripper**, bring the two exposed strap ends up around the sides of the bundle. The left gripper may slightly adjust its grip to expose the middle while keeping the loops aligned on the strap.
2. Overlap the hook-and-loop surfaces and press the closure until the strap is secure.

**Expected state:** the strap is closed around the bundle middle with its hook-and-loop surfaces overlapped, every loop is captured inside it, and the cable is not pinched or kinked.

#### **4.4 Lift the strapped bundle with the right gripper**

1. Transfer control of the strapped bundle to the **right gripper**, grasping the bundle at or next to the strap without grasping either connector.
2. Open the left gripper only after the right gripper is closed on the bundle, then lift the bundle clear of the tabletop.

**Expected state:** the strapped bundle is held in the air by the right gripper, ready to carry, and the left gripper is clear.

**Check:** the strap was laid across the middle of the table before the bundle was lowered onto it; the strap is now centered around the bundle, the closure overlaps, all loops remain gathered, and the cable is not pinched or kinked. If the strap opens or a loop slips free, lower the bundle back onto the center work area and refasten it before stowing.

### **Step 5: Stow the finished bundle and start the next cable**

**Goal:** the strapped bundle rests alone in the next empty organizer compartment, and the next cable turn begins only from that finished state.

1. With the **right gripper**, carry the secured bundle from the center to the back-right organizer. Keep both connectors clear of the tabletop and basket.
2. Identify the **next empty compartment**: leftmost on turn 1, second on turn 2, third on turn 3, and rightmost on turn 4.
3. Lower the bundle into that compartment with its loops contained inside the compartment walls.
4. Release only after the bundle is supported by the compartment. Do not move bundles stowed earlier in the episode.
5. If loose cables remain in the left input zone, return to Step 1 and run the next cable turn. Only after the fourth bundle is stowed does the episode continue to Step 6.

**Check:** the bundle remains inside the correct compartment, the strap stays closed, no connector is trapped under a compartment edge, and the center work area is clear before the next cable is selected.

### **Step 6: End the episode**

**Goal:** recording ends after all four cables have been coiled, secured, and stowed.

1. Confirm all four compartments hold one stable strapped bundle each, filled left to right, that the left input zone is empty, and that the center work area is clear.
2. Return both arms home with grippers open. Homing is the last thing the arms do.
3. Stop recording.

## **After the episode: reset the workspace**

All reset work happens with recording off.

1. Remove all four bundles from the organizer.
2. Open their straps and return the straps to the front-right basket.
3. Fully uncoil all four cables.
4. Inspect the cables, connectors, straps, basket, and organizer. Replace any frayed, broken, or cracked item.
5. Stage four loose, uncoiled cables in the left input zone with their connectors visible and not knotted together.
6. Confirm all four organizer compartments are empty, then run both Setup checklists before the next episode.

## **SOP violations**

Things that break this SOP and that reviewers look for in the side-by-side review tool.

## How to record a violation in review

For every violation seen in a recorded episode, record:

- the **start timestamp** in the video;
- the **violation name** from the list below; and
- the **SOP rule broken**, including the step number.

The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

## Episode handling

Tag every violation with its timestamp and name. An episode may contain multiple violations; tag each separately, including the same violation repeated on different cables. Retain the episode in training data with its violation tags. Do not delete a recorded episode solely because it contains a violation.

## Violations

### Violation: More than one cable handled at a time

- **Visible cue:** a second cable is pulled into the center, coiled, or moved out of the left input zone while another cable is still being handled, or the next cable turn begins before the current bundle is stowed.
- **SOP rule broken:** Steps 1 and 5 (isolate one cable, stow it, and only then start the next turn).
- **Coaching note:** one cable at a time, start to finish, before the next one leaves the left input zone.

### Violation: Incomplete cable set

- **Visible cue:** recording ends with fewer than four bundles in the organizer, or with a cable still loose in the left input zone, coiled but unstrapped, or lying in the center work area.
- **SOP rule broken:** Steps 5 and 6 (run four cable turns and end the episode only after the fourth bundle is stowed).
- **Coaching note:** four cables per episode. Finish the set before homing.

### Violation: Cable not isolated cleanly

- **Visible cue:** the selected cable is pulled through another cable, remains crossed with another cable, or drags a second cable into the center.
- **SOP rule broken:** Step 1 (lift crossings free and move only the selected cable into the center).
- **Coaching note:** stop at each crossing and separate it without pulling.

### **Violation: Remaining cables displaced from the input zone**

- **Visible cue:** a remaining loose cable is pushed into the center work area, off the front or back table edge, or under the strap basket or organizer, and is left there.
- **SOP rule broken:** Steps 1 and 6 (leave the remaining cables in the left input zone and clear that zone only by completing them).
- **Coaching note:** work the selected cable free of the group and leave the rest where they lie.

### **Violation: Cable pulled hard instead of released**

- **Visible cue:** a gripper keeps pulling on a cable that is caught, crossed with another cable, or knotted, or yanks it free, instead of releasing and freeing the catch.
- **SOP rule broken:** Steps 1 and 2 (lift a crossing free without pulling the cables apart, and release rather than pulling hard when the cable catches).
- **Coaching note:** release, find the catch, free it, then continue the pass.

### **Violation: Coiling begins before straightening is complete**

- **Visible cue:** the operator starts making loops while the selected cable still contains a knot, closed loop, crossing, or hidden connector.
- **SOP rule broken:** Step 2 (work to the free connector and confirm the entire cable is clear before forming the first loop).
- **Coaching note:** complete the full release, advance, regrasp, and straighten pass first.

### **Violation: Connector mishandled**

- **Visible cue:** a gripper closes on a connector, pulls it, crushes it, or drags it across the table during selection, coiling, fastening, or stowing.
- **SOP rule broken:** Steps 1–5 (grasp the cable body and keep connectors free).
- **Coaching note:** grip the insulated body just in from a connector, never the connector itself.

### **Violation: Wrong coil pattern**

- **Visible cue:** loops alternate onto opposite sides to form a figure eight, or the cable is wound around a gripper instead of being delivered section by section into one stacked bundle.
- **SOP rule broken:** Step 3 (form every loop in the same direction and stack each delivered section in the holding gripper).
- **Coaching note:** the hands alternate, the loop direction does not. Repeat the sample's same-direction advance, deliver, and gather motion.

### **Violation: Hands not alternated**

- **Visible cue:** the same gripper holds the bundle for two loops in a row, or one gripper delivers every loop while the other only holds, so the bundle never changes hands.
- **SOP rule broken:** Step 3 (swap the holding and delivering roles after every loop).

- **Coaching note:** one loop per gripper. Hand the bundle over before advancing for the next section.

**Violation: Bundle not finished in the left gripper**

- **Visible cue:** the coiling ends with the bundle in the right gripper and strapping begins from there, or the left gripper fetches the strap while the right gripper holds the bundle.
- **SOP rule broken:** Steps 3 and 4 (deliver the last loop with the right gripper and hand the bundle back to the left gripper before fastening).
- **Coaching note:** close the alternation on the left, then let the right gripper go for the strap.

**Violation: Bundle dropped or set down during a hand-off**

- **Visible cue:** the holding gripper opens before the receiving gripper is closed on the bundle, a loop slips free during the swap, or the bundle touches the tabletop while changing hands.
- **SOP rule broken:** Step 3 (close the receiving gripper before releasing the other, and keep the bundle in the air through every hand-off).
- **Coaching note:** grip before release, and keep the swap above the center work area.

**Violation: Bundle formed on the table**

- **Visible cue:** a formed loop or the growing bundle is laid on the tabletop and arranged there.
- **SOP rule broken:** Step 3 (form and gather the full loop bundle in the air).
- **Coaching note:** keep the growing bundle raised through every hand-off and until the open strap is laid across the middle of the table.

**Violation: Coiling or strapping done outside the center work area**

- **Visible cue:** loops are formed, handed off, or strapped over the left input zone, the strap basket, or the organizer rather than in the center work area.
- **SOP rule broken:** Steps 3 and 4 (form and hand off the bundle above the center work area, then lay out the strap and fasten the bundle on the table there).
- **Coaching note:** bring the cable to the center first, keep every loop and hand-off over it, and perform the full strapping sequence there.

**Violation: Uneven or incomplete bundle**

- **Visible cue:** one loop projects far beyond the others, a loop is kinked or twisted, or a long uncoiled tail remains when fastening begins.
- **SOP rule broken:** Step 3 (make even loops through the free connector and correct a bad loop before fastening).
- **Coaching note:** match the first loop size and continue to the free end.

**Violation: Bundle too wide for a compartment**

- **Visible cue:** the finished bundle is visibly wider than one compartment outline, and it is strapped and carried to the organizer without being remade at a smaller loop size.
- **SOP rule broken:** Steps 3 and 5 (make compartment-sized loops, and remake an oversized bundle before fastening and stowing).
- **Coaching note:** set the loop size on the first loop and match it, so the bundle drops inside one compartment.

#### **Violation: Strap not staged before bundle placement**

- **Visible cue:** the coiled cable is lowered onto the bare table before the strap is laid out, the strap is placed away from the middle of the table, or the strap is slid under a bundle that is already resting on the table.
- **SOP rule broken:** Step 4 (lay one open strap across the middle of the table first, then place the coiled cable on top of it).
- **Coaching note:** strap on the middle of the table first; coiled cable on top second; then fasten.

#### **Violation: No strap, or more than one strap used**

- **Visible cue:** a bundle is carried to the organizer with no strap on it, or a second strap is added to the same bundle.
- **SOP rule broken:** Step 4 (close exactly one supply strap around the bundle middle).
- **Coaching note:** one strap per bundle, taken from the front-right basket.

#### **Violation: Strap off the bundle middle or closure not overlapped**

- **Visible cue:** the strap sits away from the bundle middle, the hook-and-loop surfaces do not overlap, or a loop remains outside the closed strap.
- **SOP rule broken:** Step 4 (close the strap around the bundle middle with its surfaces overlapped and every loop captured).
- **Coaching note:** wrap at the middle, capture every loop, and press the overlap closed.

#### **Violation: Cable pinched or kinked by the strap**

- **Visible cue:** the closed strap visibly pinches, flattens, or sharply bends the cable.
- **SOP rule broken:** Step 4 (secure the strap without crushing or sharply bending the cable).
- **Coaching note:** close the strap only far enough to keep the loops together.

#### **Violation: Unsecured bundle stowed**

- **Visible cue:** the strap opens or a loop slips free and the bundle is still moved into the organizer.
- **SOP rule broken:** Steps 4 and 5 (refasten a loose bundle before carrying it to the organizer).

- **Coaching note:** correct the strap over the table before stowing.

#### **Violation: Bundle carried over the basket or dragged on the table**

- **Visible cue:** the strapped bundle or a connector scrapes the tabletop, drops into or drags across the strap basket, or swings against the organizer while being carried.
- **SOP rule broken:** Step 5 (carry the bundle with both connectors clear of the tabletop and the basket).
- **Coaching note:** lift clear, carry level, and cross above the basket rather than through it.

#### **Violation: Wrong stow compartment or placement**

- **Visible cue:** a bundle is placed outside the organizer, skips an empty compartment, shares a compartment, projects over a compartment wall, or disturbs a bundle stowed earlier in the episode.
- **SOP rule broken:** Step 5 (place each bundle alone in the next empty compartment from left to right and leave earlier bundles undisturbed).
- **Coaching note:** use the first empty compartment from the left and lower the bundle fully inside.

#### **Violation: Organizer or strap basket displaced**

- **Visible cue:** a gripper, an arm, or the carried bundle pushes, slides, or tips the organizer or the strap basket, or knocks an earlier bundle out of its compartment.
- **SOP rule broken:** Steps 4 and 5 (take a strap and stow the bundle without disturbing the basket, the organizer, or bundles completed earlier in the episode).
- **Coaching note:** approach both from above on a clear path and keep the arm off their walls.

#### **Violation: Wrong arm used**

- **Visible cue:** outside Step 3, an action assigned to one gripper is performed by the other, including the left gripper fetching the strap or the left gripper carrying a finished bundle to the organizer.
- **SOP rule broken:** Steps 1, 2, 4, and 5 (use the left gripper to hold, and the right gripper to straighten, lay out and fasten the strap, and stow). Step 3 is the exception: there the grippers alternate by design, and a role swap is scored under **Hands not alternated** instead.
- **Coaching note:** the hands alternate only while loops are being formed. Everywhere else the left gripper is the bundle support and the right gripper is the working and stowing gripper.

#### **Violation: Wrong episode ending**

- **Visible cue:** recording stops before the fourth bundle is stowed and the end state is confirmed, an arm is not home, a gripper is closed, or an arm performs another action after homing.

- **SOP rule broken:** Step 6 (confirm the completed state, return both arms home with grippers open as their final action, then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arms do.

### **Non-violation failures**

Failures not caused by task execution are system issues. Log and discard the episode rather than tagging them as SOP violations.

- Recording stops or pauses during the episode.
- A camera drops frames or loses its feed.
- An arm or gripper fails, drifts, or reports a motor error.
- A cable, connector, strap, basket, or organizer is defective or damaged.

### **Annotation subtasks (from SOP)**

1. Select and isolate one cable from the left input zone
2. Straighten the cable between the grippers
3. Form a compact same-direction loop bundle in the air, alternating the grippers on every loop
4. Lay one strap across the middle of the table, place the coiled cable on top, and secure it
5. Stow the secured bundle in the next empty organizer compartment
6. Repeat subtasks 1-5 for the remaining cables until all four compartments are filled
7. Confirm the end state, return both arms home, and end the episode